

HR Attrition Analysis

Uncovering patterns in employee turnover using SQL & BigQuery

Tools: BigQuery · SQL · Google Data Studio

Portfolio Project · Refia Ayse Ergelen

Project Overview

Objective

Identify which departments, roles, and salary bands drive the highest employee attrition.

Dataset

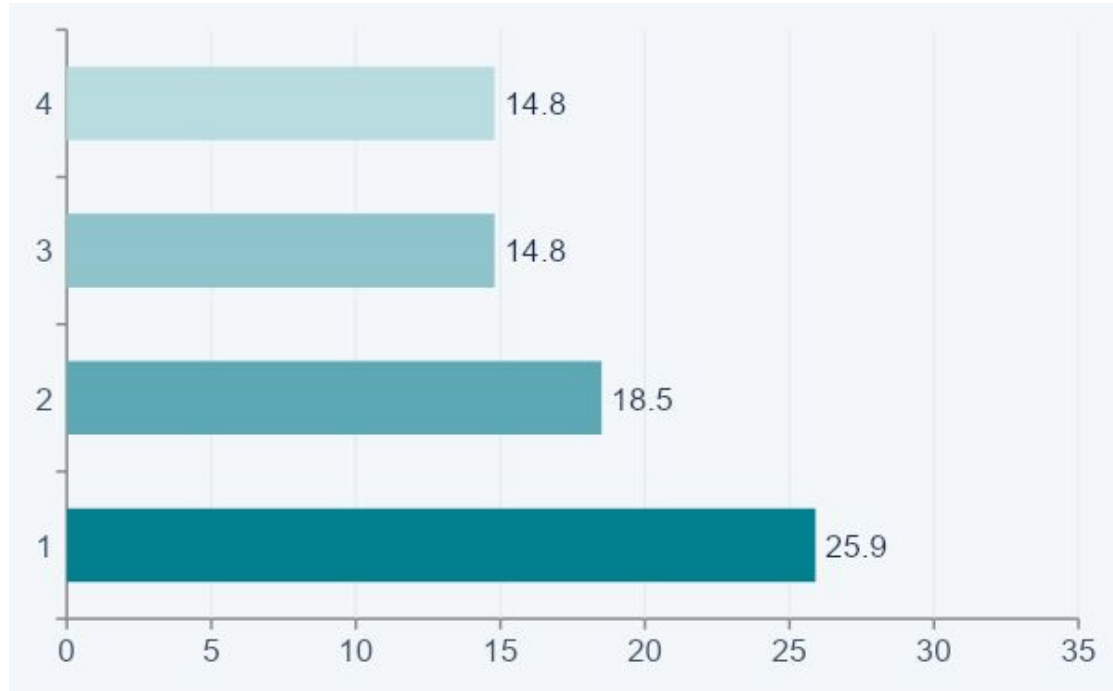
HR records with employee tenure, salary, department, job title, hire/exit dates, and attrition flags.

Approach

Data cleaning in BigQuery views, then layered SQL analysis: attrition rates, salary benchmarking, and tenure segmentation.

Key Finding 1

Customer Support Department Leads Attrition



25.9%

of all attrition
from Customer Support

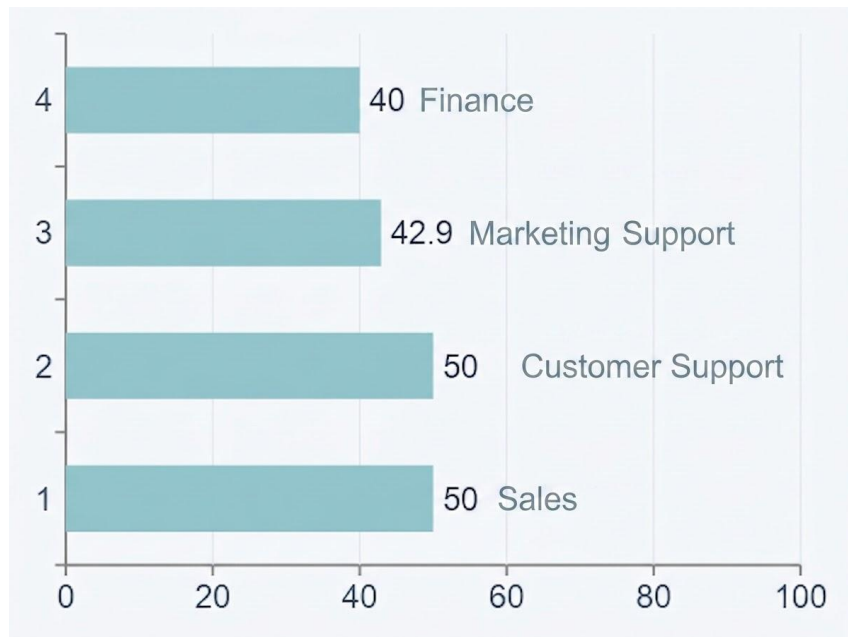
Business Takeaway

Nearly double the rate of Sales and Marketing, Customer Support is worth investigating for retention.

Key Finding 2

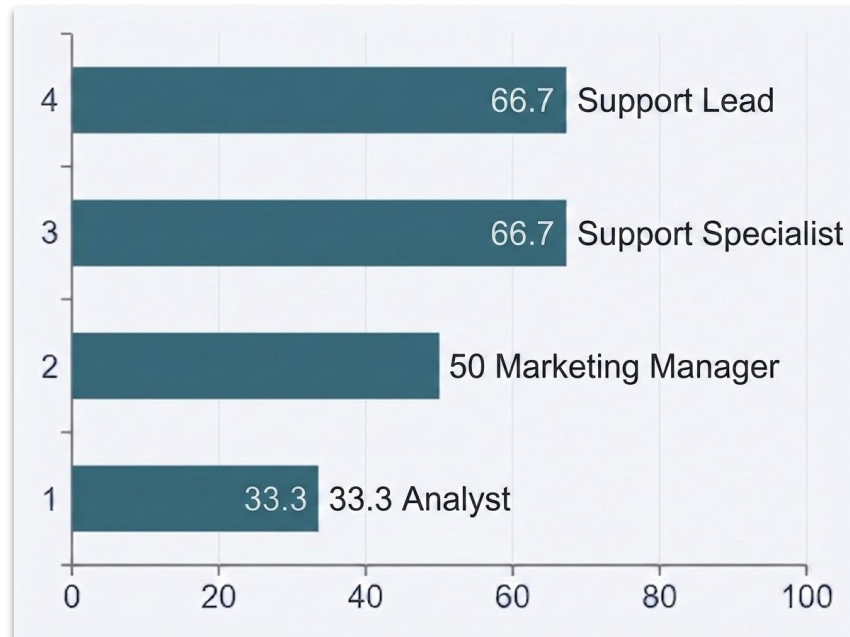
Salary Based Attrition is Role-Specific, Not Universal

By Department



% of attrition below average department salary

By Job Title



% of attrition below average job title salary

Key Finding #2 Business Takeaway

- Department-level data looks even (40–50%) suggesting attrition isn't a function of being underpaid relative to your department. But at the job title level, Support Lead and Support Specialist hit 66.7%, pointing to a compensation issue in specific support roles.
- Two-thirds of Support Leads and Support Specialists who left were paid below their peer average.
- Conduct targeted pay equity review for support roles specifically, instead of a company-wide salary adjustments. Consequently, you'd likely see a big drop in the highest attrition department, Customer Support.

Key Finding 3

Prioritize manager retention



Major Costs

Manager exits disrupt team direction and slow project delivery. Departing managers often trigger further exits among their direct reports.

Business Takeaway

The data doesn't point to an internal salary gap for this role, so it's worth checking whether Marketing Manager pay is competitive against market rate. From there, either adjust pay bands or build a support plan for managers

Summary – Recommendations

01

Support role compensation

Support Lead and Support Specialist show 66.7% below-average salary attrition. A targeted pay review is crucial.

02

Invest in Customer Support retention

At 25.9% of all attrition, this department needs structural intervention, not just perks.

03

Protect manager continuity

Marketing Manager attrition at 14.8% carries major downstream risk. Support planning and manager engagement surveys are high-ROI.

Methodology

- 1 Data cleaning**
Standardized date formats, salary strings, department labels, and location codes into a BigQuery view.
- 2 Tenure segmentation**
Classified exits as New Hire Churn (<90d), Early Attrition, Mid-Term, or Veteran Attrition using DATE_DIFF.
- 3 Attrition rate queries**
Calculated % share of total attrition by department and job title using CTEs and window functions.
- 4 Salary benchmarking**
Used PARTITION BY to compare each departing employee's salary to their department and job title average.